

NEUCHATEL, Switzerland

WMO: 066040

Lat: 47.0000N

Lon: 6.9533E

Elev: 487

StdP: 95.61

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -6.3	(c) -4.7	(d) -10.8	(e) 1.6	(f) -4.2	(g) -8.9	(h) 1.9	(i) -2.8	(j) 10.5	(k) 4.1	(l) 9.4	(m) 3.8	(n) 3.0	(o) 70	(p) 0.472

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.9	(c) 29.9	(d) 19.6	(e) 28.0	(f) 19.0	(g) 26.3	(h) 18.3	(i) 20.5	(j) 27.6	(k) 19.7	(l) 26.2	(m) 19.0	(n) 24.9	(o) 2.3	(p) 190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.1	13.8	22.9	17.4	13.2	22.3	16.8	12.7	21.7	60.9	27.8	58.2	26.4	55.9	25.0	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.7	(b) 7.5	(c) 6.5	(e) -8.1	(f) 32.4	(g) 2.2	(h) 2.0	(i) -9.7	(j) 33.9	(k) -11.0	(l) 35.1	(m) -12.2	(n) 36.2	(o) -13.9	(p) 37.6		
(4) 8.7	(5) 7.5	(6) 6.5	(e) -8.8	(f) 22.0	(g) 2.2	(h) 0.8	(i) -10.4	(j) 22.5	(k) -11.6	(l) 23.0	(m) -12.9	(n) 23.4	(o) -14.5	(p) 24.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	1.8	2.8	6.7	10.5	14.5	18.4	20.3	19.8	15.6	11.3	5.9	2.6
	DBStd	7.43	3.67	3.80	3.64	3.68	3.39	3.70	3.26	3.20	3.11	3.29	3.37	3.52
	HDD10.0	992	254	201	112	38	4	0	0	0	1	25	128	229
	HDD18.3	2920	512	434	360	236	127	44	17	21	91	218	373	487
	CDD10.0	1322	0	1	10	53	143	252	320	303	170	65	4	0
	CDD18.3	208	0	0	0	0	7	47	79	66	10	0	0	0
	CDH23.3	1284	0	0	0	2	39	291	519	398	35	0	0	0
	CDH26.7	315	0	0	0	0	3	65	141	105	1	0	0	0
Wind	WSAvg	2.4	2.6	2.8	2.8	2.5	2.4	2.3	2.4	2.1	2.3	2.0	2.3	2.6
Precipitation	PrecAvg	978	76	69	69	65	76	104	86	114	85	53	104	75
	PrecMax	1340	164	225	135	108	110	191	202	212	196	131	185	198
	PrecMin	741	24	11	8	17	36	34	33	33	18	3	34	2
	PrecStd	151	36	48	36	31	22	50	47	56	56	33	47	56
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.3	13.0	18.0	23.1	26.5	31.0	32.3	32.2	26.4	20.8	15.2	11.3
		MCWB	8.3	8.3	10.5	13.4	17.2	20.5	19.8	20.1	18.0	14.9	11.3	8.6
	2%	DB	9.7	10.7	16.0	20.7	24.2	28.6	30.0	29.4	24.2	18.6	13.0	10.0
		MCWB	7.4	6.9	9.5	12.4	16.0	19.1	19.6	19.6	17.5	14.2	10.3	8.0
	5%	DB	8.2	9.2	14.0	18.7	22.4	26.6	28.1	27.3	22.4	17.0	11.7	8.8
		MCWB	6.2	6.2	8.5	11.5	14.8	18.4	18.9	19.0	16.6	13.7	9.5	7.0
	10%	DB	6.9	7.8	12.2	16.7	20.5	24.7	26.2	25.4	20.6	15.7	10.4	7.5
		MCWB	5.2	5.3	7.8	10.6	14.0	17.5	18.3	18.2	15.7	13.0	8.6	6.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.2	8.7	11.4	14.5	18.2	21.4	21.3	21.3	19.3	16.1	12.0	9.4
		MCDB	10.6	12.0	16.4	21.4	24.8	28.8	29.2	28.6	24.5	18.8	13.9	10.5
	2%	WB	7.6	7.4	10.2	13.0	16.9	20.0	20.3	20.3	18.1	15.1	10.8	8.3
		MCDB	9.3	10.1	14.3	19.3	22.7	26.8	27.7	27.2	22.5	17.6	12.6	9.6
	5%	WB	6.5	6.5	9.3	12.0	15.7	19.1	19.6	19.6	17.3	14.2	9.8	7.2
		MCDB	8.1	8.9	13.1	17.5	20.8	25.1	26.4	25.7	21.3	16.3	11.3	8.6
	10%	WB	5.3	5.5	8.2	11.1	14.6	18.1	18.9	18.9	16.3	13.3	8.9	6.1
		MCDB	6.7	7.7	11.8	15.9	19.4	23.7	25.0	24.3	19.8	15.3	10.2	7.3
Mean Daily Temperature Range	5% DB	MDBR	3.6	5.0	7.0	8.0	8.2	8.6	8.9	8.4	7.2	5.5	4.0	3.5
		MCDBR	5.2	7.1	10.1	11.3	11.3	11.1	11.7	11.4	9.5	7.1	5.6	4.8
		MCWBR	3.9	4.4	5.2	5.3	5.4	5.0	4.8	4.7	4.6	4.1	4.0	3.7
	5% WB	MCDBR	5.0	6.3	8.7	10.3	10.1	10.4	10.5	10.1	8.5	6.3	5.2	4.5
		MCWBR	3.9	4.3	4.8	5.1	5.0	4.9	4.6	4.5	4.4	3.8	3.9	3.7
Clear-Sky Solar Irradiance	taub	0.295	0.320	0.357	0.395	0.404	0.406	0.405	0.397	0.375	0.354	0.320	0.293	
	taud	2.438	2.373	2.331	2.252	2.271	2.306	2.303	2.338	2.379	2.424	2.464	2.447	
	Ebn at Noon	800	845	862	858	861	860	855	847	833	795	760	757	
	Edh at Noon	67	88	107	127	130	126	125	116	102	84	66	60	
All-Sky Solar Radiation	RadAvg	1.23	2.07	3.25	4.36	5.02	5.94	5.76	5.01	3.87	2.46	1.35	0.99	
	RadStd	0.15	0.31	0.48	0.64	0.47	0.55	0.51	0.46	0.35	0.24	0.17	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (105 neighbors)		N/A	N/A	N/A	+0.73	+0.40	+0.30	N/A	N/A	+53	+17

Nomenclature: See separate page